Unit II

Work with Machines and Materials

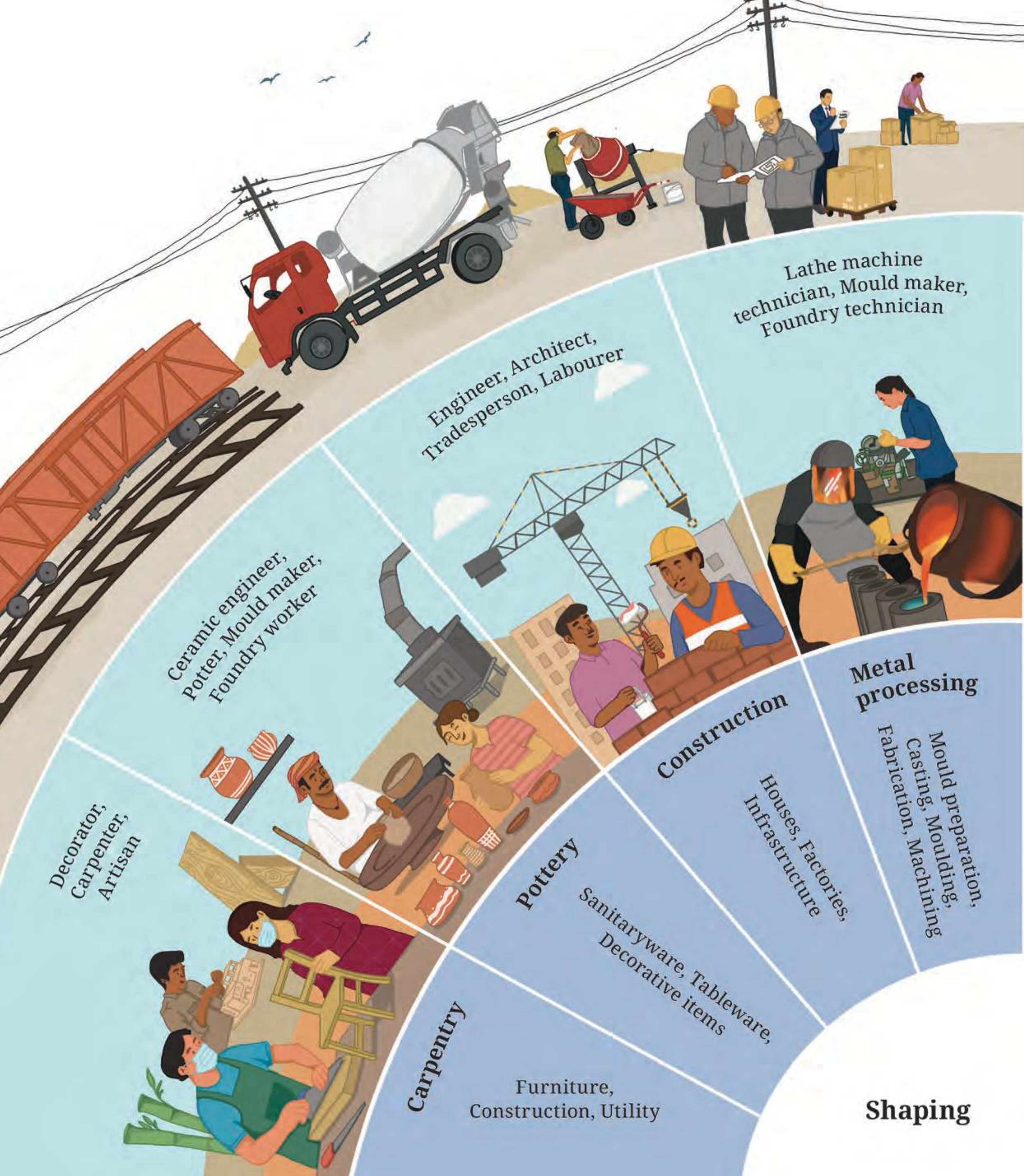
From the time humans first shaped a stone into a tool, our progress has been closely linked with our ability to use materials and build machines. Every object around us – a clay pot, a wooden chair, a bicycle or a computer – tells a story of how we have learnt to transform raw materials from nature into useful products.

Working with machines and materials means understanding both the characteristics of materials, and the tools and processes used to shape them. Whether it is forging iron, weaving fabric or building a bridge, each activity combines knowledge, skill and creativity.

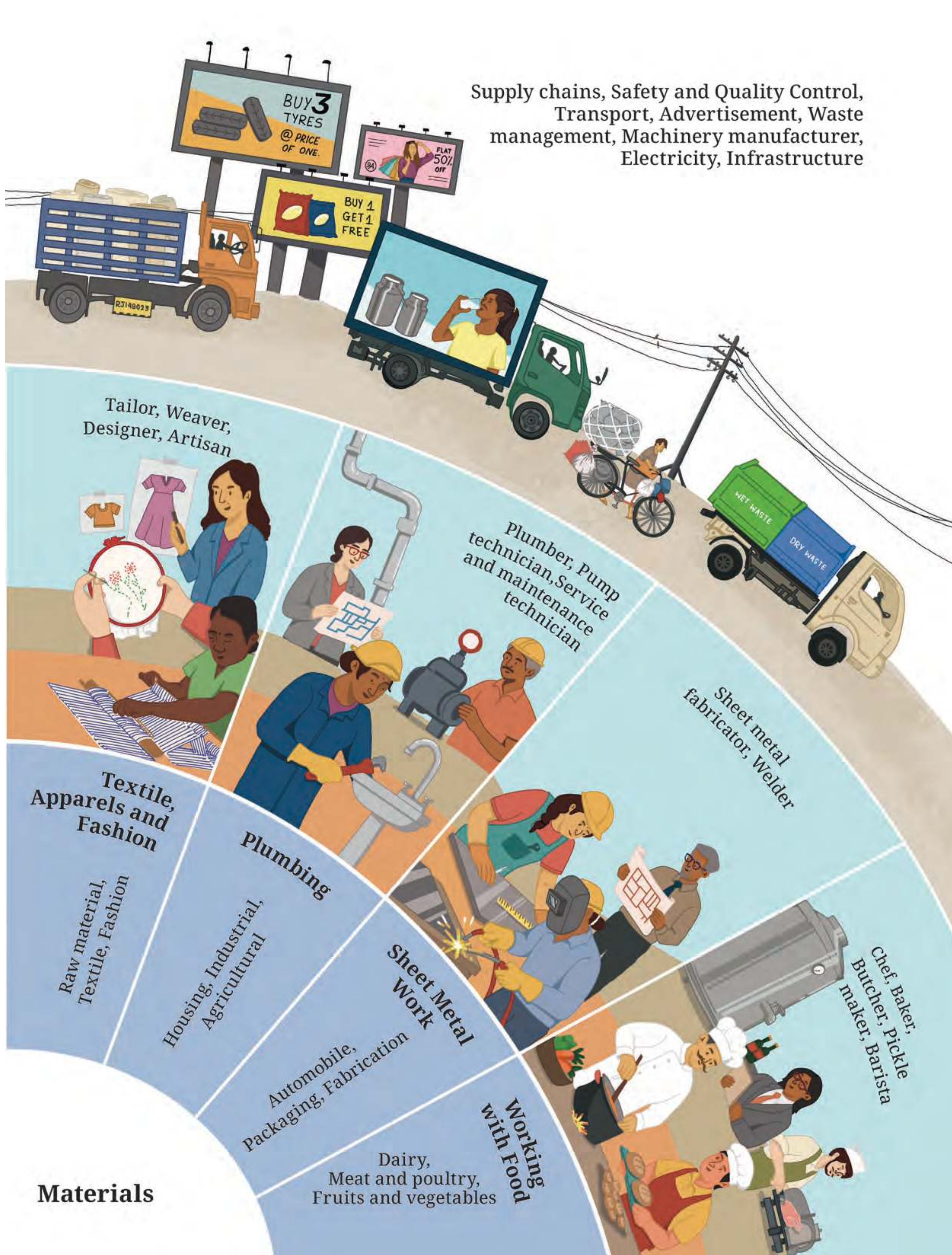
In ancient India, people showed remarkable craftsmanship – the Iron Pillar of Delhi, the rock-cut caves of Ellora and the textiles of the Sindhu – Sarasvatī civilisation – all reflect deep understanding of materials and techniques. Today's industries are built upon the foundation of these traditions.

In today's world, technological advances are not only increasing efficiency but also protecting workers. 'Smart systems' are being developed that can warn workers of potential accidents and enable them to take preventive measures. For example, 'smart' helmets and vests are equipped with sensors that monitor heart rate, body temperature and even posture. Thus, if a welder's body temperature rises to a dangerous level, or a technician's spine and limbs are placed at dangerous angles, a warning will be given. Technology is also taking over jobs that are dangerous, for example, climbing up a tower to check for structural cracks or checking areas for gas leaks or structural damage.

With the arrival of machines and industrialisation, production became faster and more precise. Modern tools, automation and digital technology now help us design and make things more efficiently than ever before. Yet, the basic principles remain the same – selecting the right material based on its properties, using the right process and handling tools and machines safely and skilfully.



Lathe machine technician, Mould maker, Foundry technician
Engineer, Architect, Tradesperson, Labourer
Ceramic engineer, Potter, Mould maker, Foundry worker
Decorator, Carpenter, Artisan
Metal processing
Mould preparation, Casting, Moulding, Fabrication, Machining
Construction
Houses, Factories, Infrastructure
Pottery
Sanitaryware, Tableware, Decorative items
Carpentry
Furniture, Construction, Utility
Shaping



Supply chains, Safety and Quality Control, Transport, Advertisement, Waste management, Machinery manufacturer, Electricity, Infrastructure
BUY 3 TYRES @ PRICE OF ONE.
FLAT 50% OFF
BUY 1 GET 1 FREE
WET WASTE
DRY WASTE
Tailor, Weaver, Designer, Artisan
Plumber, Pump technician, Service and maintenance technician
Sheet metal fabricator, Welder
Chef, Baker, Butcher, Pickle maker, Barista
Textile, Apparels and Fashion
Raw material, Textile, Fashion
Plumbing
Housing, Industrial, Agricultural
Sheet Metal Work
Automobile, Packaging, Fabrication
Working with Food
Dairy, Meat and poultry, Fruits and vegetables
Materials

The figure overleaf depicts the livelihood ecosystem related to machines and materials. A livelihood ecosystem is an interconnected network of resources, people, institutions, activities and environmental factors that enable individuals to earn a living, while contributing to society and nation. For instance, the livelihood ecosystem related to machines and materials varies across geographies, depending on the availability of raw materials (like metal and wood) access to machines and tools (manual, electrical or automated); support services (like technicians and repair workers); infrastructure (like electricity) access to markets (like transportation) and the demand for manufactured goods at local, national or global levels. Further, this implies that no work is done in isolation–different kinds of work are deeply connected. For example, if demand for a particular kind of product reduces, then the manufacturer will not be the only one affected. Slowly, requirement for transport, production of agricultural equipment and of technicians will also decrease. As you can see from the figure, there are a range of opportunities related to working with machines and materials. Besides construction and manufacturing, the vocational area includes working with wood, bamboo and food as well as apparels, fashion, textiles, plumbing, pottery and sheet metal work; the list is long. Each level in the figure provides details of the kind of work that can be done in this area. The last level indicates the interlinkages between the work in the area and various other allied work that enable working with machines and materials, while ensuring that society benefits.

In this chapter, you will explore how materials are selected, how they can be measured, shaped and joined, and how machines make this transformation possible. It will help you trace a journey–from matter to machine and from idea to creation.

This unit will give you the opportunity to do work related to machines and materials. It is called 'Shaping materials' because you will select materials based on their properties and process them into useful products. This unit offers illustrations of seven vocations, as explained in the paragraphs below.

Chapter 5 introduces key concepts and processes that are common across a range of work related to machines and materials. This chapter is mandatory for you. Chapters 6 to 8 use the common concepts and processes to help you understand how to do specific work. At the same time, they also introduce additional concepts and processes that you can learn, while doing. Chapters 6 and 7 detail work related to construction and apparel, respectively. Chapter 8 contains guidelines for sheet metal work, plumbing, food processing, furniture making and pottery.

To reiterate, you can choose to do work related to any of these seven vocations or you can select something related to shaping materials that is entirely different. Remember to consult your teacher and/or an expert for guidance at all points. Work is to be done in groups. Remember–a big part of work is doing it together.

CHAPTER 5

Shaping Materials

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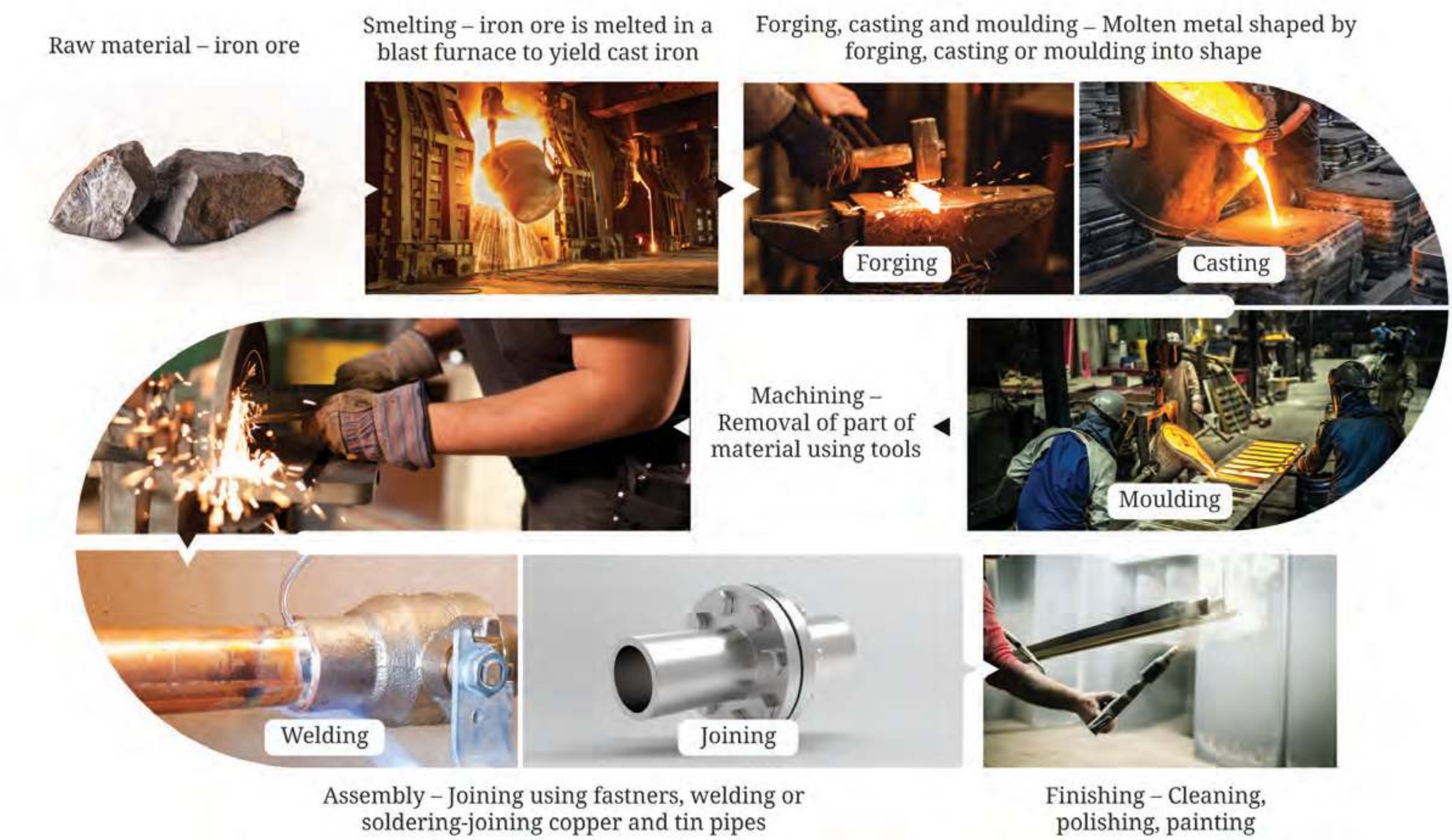


Figure 5.1: *Materials are shaped into products through different processes*

All materials come from nature, and we shape them into different forms. The process involves gathering raw materials and manufacturing them into finished products through various processes (Figure 5.1).

In this chapter, you will

Explore different kinds of materials and their characteristics	Understand common processes related to shaping materials	Identify different tools for measurement and their applications
Select materials based on their characteristics	Understand safety protocols	Read technical drawing accurately
Create simple technical drawings	Understand quality criteria	Explore vocations related to shaping materials

Importance of work

5.1 Introduction

India has a rich history of shaping materials through various processes to suit specific needs. To illustrate, evidence from the Sindhu–Sarasvatī civilisation reveals that houses were built with bricks of the same size. Harappans mass produced bricks of a standard size, ensuring perfect fit and uniform strength. This involved moulding the wet clay and then controlling the firing process (curing) in large kilns to achieve the hardness and durability required for building. Besides bricks, intricate copper and brass objects, like the famous 'Dancing Girl' were created using casting.

Evidence also exists of the knowledge of working with iron. For example, a pillar made of purified iron was created during the fourth century CE. This pillar, which now stands in the Qutub Minar complex in Delhi, still has not rusted as this natural property of iron has been changed due to a protective layer.

The Kailāśha Temple in Ellora; the Koṇārka temple in Puri, Odisha; the Bṛihadīśhvara Temple in Thanjavur; and many such temples across India stand testament to the skills of the engineers who designed and constructed them, their patience and hard work. Their work shows evidence of deep practical craftsmanship, and an understanding of concepts related to metallurgy, architecture, and engineering.

In the more recent past, carpenters, blacksmiths, weavers, tailors, masons, cobblers, oil pressers, potters, jewellers and others were present in every village. These professionals learnt about the characteristics of materials and how to shape them, using practical knowledge of metallurgy, construction, textiles, soil, etc., as well as the use of tools for making agricultural implements, pots, vessels, jewellery, weapons, etc. Manufacturing in earlier times solely depended on the skills of individuals being passed on from generation to generation.

DID YOU KNOW?

Materials with unexpected characteristics

Our planet is truly remarkable. It is filled with materials, each one with fascinating characteristics that are useful in countless ways. Some materials have unusual characteristics or are procured from unexpected sources.

Glass from sand: The transparent, hard material used in windows and jars is made simply by heating fine sand to very high temperatures until it melts.

Obsidian: It is a naturally occurring glass formed from the lava extruding from volcanoes. A blade made from obsidian can be 500 times sharper than a surgeon's steel scalpel.

Artificial spider silk: Natural spider silk is five times stronger than steel by weight. Scientists have successfully engineered a way to create artificial spider silk using bacteria fed on sugar, creating a biodegradable material with immense potential for advanced textiles (such as parachute fabric and protective clothing) and industrial applications.

After industrialisation, many manual tools were replaced by electric power tools. This increased the efficiency of processes, and the pace and quality of production. As a result, traditional jobs and skills were replaced with new kinds of work, while some basic skills, like measurement, design, safety precautions and testing products for quality persisted.

The manufacturing sector contributes about 17–18 per cent of India's Gross Value Added (Ministry of Statistics and Programme Implementation, National Accounts Statistics 2025) and remains an important source of employment.

Importance of work

Value addition

Whenever we transform materials from their natural form to useful products through various processes, their value increases. For example, the value of a simple stone lying on the hillside increases manifold when a craftsman chisels a statue out of it.

Selling a material in its natural form costs less than a finished product. As shown in the example of cotton (Figure 5.2), if we convert cotton to fabric and fabric to garment, then its value increases substantially.

Value chain of work

Value chain of work

Figure 5.2: *Value chain of cotton*

TASK
List all the factors that add value to any raw material. Ask around, search in the library, search online, read newspapers and identify experts who can help you.

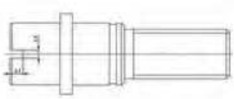
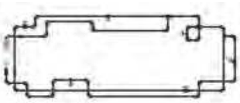
Raw to final product	Value added
Metal scrap (₹ 20/kg) to kitchen utensil (₹ 1200)	Undergoes processing through which it is melted and moulded to form the shape of the utensil. It is polished or coated with anti-sticking materials to improve its quality. Wooden handles are attached for better handling of the utensil. The bottom of the utensil can be further processed, so that it is both stove and induction-friendly.

Explore different sources of information

5.2 Processing material – transformation to a product

Key processes for shaping materials

Different manufacturing processes shape raw materials into useful products. These processes have evolved over thousands of years, and become more and more specialised. However, some processes are common across a range of materials, for example, fabric, cloth, leather, metals and plastics as illustrated in Figure 5.3.

Design and Estimate	Measure and mark where material is to be added or removed	Cut and shape	Join and Assemble	Finish
Fabric				
				
Use pencil to draw pattern	Mark pattern on fabric with chalk	Cut using scissors	Use thread and needle, or sewing machine	Ironing, trimming
Wood				
				
Draw product to be made on paper	Mark cuts and drilling points	Cut with a saw, wood held with a vise	Use adhesive, or hammer and nails	Sanding, polishing and painting
Metal				
				
Technical drawing/CAD	Mark with an engraving tool	Use cutting/ power tools	Use bolts and rivets, welding, soldering, etc.	Finish using grinder and paint
Plumbing				
				
Layout diagram of piping	Mark cuts on pipe and join points	Cut pipes using PVC cutter/ hacksaw	Use brazing, coupling, adhesives, etc.	Check for leakage, tighten joints

Construction				
				
Draw a sketch of the construction	Mark area using marking powder/chalk	Level site lay foundation	Lay bricks, apply mortar	Plastering, painting, flooring, electrification
Safety – Wear protective gear and follow safety rules				
				
Organise work area	Measure accurately without damaging the material	Follow all safety protocols	Handle adhesives and fasteners with care	Keep working area clean

Figure 5.3: *Common processes involved in transforming materials into products*

TASK

Note different products in your surroundings. For example, utensils in the kitchen, screw, plastic bucket, shoes, cup and saucers, bed, carpet or mat, electric wires, fan, automobile engine, etc.

Find out the raw materials used to make these products and the processes used to transform these materials into products.

Selection of material based on its characteristics

5.3 Choosing material for a specific product

The characteristics of a material determine whether it is suitable for certain uses. For example, we use steel or aluminium to make kitchen utensils and not plastic, as steel/aluminium can tolerate high heat and is safe for cooking, while plastic melts when heated and is therefore unsafe. Thus, selecting the right material is critical. Sturdy objects like a stool need a material that can hold weight, such as wood or metal, whereas a reusable shopping bag requires something flexible and light like jute fabric. Environmental/weather conditions also play an important role in selecting a product. You may have noticed that storage units or shelves in your bathrooms are often made from steel, PVC or aluminium, and not wood. Can you guess why?

In fact, a single material can be used for different purposes by utilising their specific characteristics. For example, clay, when soft and moist can be used for making craft items. Clay can also be moulded, dried and fired to make durable objects, like pots and bricks.

Selection of material based on its characteristics

TASK

You have studied the characteristics of materials in Science. Work in a group to pick one product each from the options provided in Table 5.1, and discuss which material would be most suitable for making it. Justify your response on the basis of characteristics of materials.

Table 5.1: Selection of materials based on their characteristics

List of products	**Best suited material for the product** (steel, copper, glass, aluminium foil, rubber, clay, fabric and wood)	**Identify the characteristics of the material that make it suitable for the product** (hardness, elasticity, thermal conductivity, electrical conductivity, plasticity, corrosion resistance, permeability, ductility, water resistance and transparency)
Raincoat		
Electric wire		
Water bottle		
Cooking pot		
Cushion		
Window		
Keychain		
School bell		

TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

Machines, Automation and Robotics

In the past, many manufacturing tasks were done manually, relying on skills of manual labour. Speed of completion of the task was slow, since it depended on the capacities of individuals. After industrialisation, power tools replaced manually-operated machines. This has increased the speed of production. Due to advances in technology, many repetitive tasks have been automated. Automated machines and robots are also more accurate and precise than humans. Such CNC (Computerised Numerical Control) machines and robots have enabled production in larger quantities.

Following safety protocols

5.4 Key common processes

Figure 5.3 in Section 5.2 contains some common processes used to create products from different materials. While the tools and materials used were different in some cases, the key processes were similar. This is possible because there are some essential skills that are common to any kind of work with machines and materials.

In this section, you will learn about three basic skills:

1. Following safety protocols.
2. Taking measurements using specific instruments.
3. Making technical drawings.

You will learn the basics of these skills and practise them in the vocation you choose.

5.4.1 Following safety protocol

Before beginning any work, read available manuals or instruction guides, and speak to your teacher and experts to understand any safety precautions that need to be taken.

Safety signage

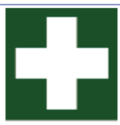
Safety signage refers to visual indicators that use standardised colours, symbols and text to communicate potential hazards, required actions, the location of safety equipment, and emergency procedures in workplaces and public areas. The purpose of safety signs is to prevent accidents and injuries by providing clear, easy-to-understand information.

Table 5.2 shows what different colours in safety signage indicate, while Table 5.3 shows examples of common safety symbols.

Table 5.2: Colours in safety signage

Colours	Meaning	Examples
Red	Fire, Prohibition	Fire extinguisher, stop buttons
Yellow	Warning and physical hazards	Wet floor caution, construction site
Blue	Mandatory actions	Wear helmet, safety instruction of a machine
Green	Providing guidance/safe condition	Emergency exit rows, assembly point

Table 5.3: Safety symbols used in public places, schools and surroundings

Symbols	Purpose	Symbols	Purpose
	Warns of risk of electric shock		Indicating path for exit, especially during emergencies
	Indicates the location of a fire extinguisher for emergency use		Protective footwear to prevent foot injuries
	Shows that helmet must be worn to protect head from any heavy falling object	 SMOKING PROHIBITED	No smoking zone to prevent fire hazards and health safety
 FIRST AID	Location of first-aid supplies or support		Warns that the floor becomes slippery when water or other liquids are present

CHECK YOUR UNDERSTANDING

Safety Audit

Do you see any of the safety signs and symbols in your surroundings? Carry out a safety check of your school and nearby areas. Discuss where these symbols should be placed and why. If you feel the need, make up your own symbol and assign it a colour. Justify its importance.

Measurement

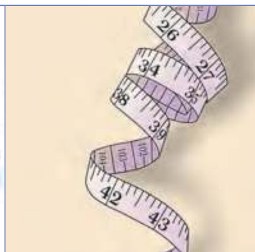
5.4.2 Taking measurements

Measurement is among the most important and basic skills for any work with materials. While you have already studied measurement in Science in Grade 6 to 8, in the context of work, measurement is needed for the following:

1. To estimate the exact amount of material required for making a product.
2. To ensure minimal wastage of material.
3. To reduce material costs and ensure the product meets the required dimensions and quality standards.

Selection of an instrument while working with materials mostly depends on how accurate the measurement needs to be (Table 5.4.) The smallest value that can be measured accurately using the instrument is called its least count. For example, the smallest measurement possible with a metre scale is 1 mm. However, the least count of a vernier callipers can be 0.02 mm, which means that it can be used to measure more precisely than a metre scale. For example, an object that measures 2.3 cm using a metre scale can be measured more precisely using a vernier callipers, that is 2.34 cm.

Table 5.4: Measuring tools and their use

				
Vernier callipers	Metre scale	Metre tape	Surveyor's tape	Distance metre
Measure diameter or thickness of small objects	Measure objects less than a metre	Measure large objects, including with curved surfaces	Measure land	Measure long distances accurately and quickly
Least count – 0.02 mm	Least count – 1 mm	Least count – 1 mm	Least count – 1 mm	Least count – 2 mm

Measurement range (Length) – 150 mm to 300 mm (Short)	Measurement range (Length) – Up to 1 m (Medium)	Measurement range (Length) – 3 m to 10 m (Medium to long)	Measurement range (Length) – 30 m to 100 m (Very long)	Measurement range (Length) – 50 m to 200 m (Long)
Required precision (Tolerance) – Very high (e.g., ±0.05 mm)	Required precision (Tolerance) – High to medium (e.g., ±0.5 mm)	Required precision (Tolerance) – Medium (e.g., ±1 mm to ±2 mm)	Required precision (Tolerance) – Low to medium (e.g., ±5 cm to ±10 cm)	Required precision (Tolerance) – Medium to high (e.g., ±1 mm over a distance)

While least count indicates the smallest measurement possible with an instrument, tolerance is the amount of error that can be allowed in a measurement. For example, if two pipes need to be joined, then the tolerance for error in measuring diameter is very low. But when measuring cloth for cutting a bag, tolerance is higher.

Therefore, selection of an appropriate measuring instrument is important. For example, to measure land, we use a 'Surveyor's tape', but to take measurement of clothes, we use a 'metre tape', or to measure the weight of a truck, a weigh bridge is used, but a grocery shop owner uses a simple balance. In a laboratory, we use a scientific balance for greater accuracy.

QUALITY

It is necessary to measure material accurately. For this, it is necessary to reduce the errors in measurement.

The following care must be taken for accurate measurement:

1. Select the instrument based on its least count.
2. Handle the instrument with care to maintain its accuracy.
3. Take due precautions specific to the material and purpose of measurement.

Quality

TASK

Select any instrument for measurement. Note the least count, the smallest measurement that can be taken by the instrument and any precaution to be taken. Two examples are given for your reference in Table 5.5.

Table 5.5: Precautions to be taken while measuring materials

Sr No.	Application	Measuring instrument	Smallest unit of measurement	Precaution
1	Measurement of cloth	Metre tape	1 mm	Keep the cloth straight without any slag during measurement.
2	Water for construction	Bucket of 15 L	1 L	Calibrate the bucket accurately.
3				
4				
5				

Technical drawing

5.4.3 Making technical drawings

Technical drawings include details, such as dimensions of different parts and specifications of materials, enabling precise communication among designers, engineers and fabricators. This allows identical replication of products in large numbers without deviations, for example, a set of dining chairs must be of the same dimension and shape; plumbing in a set of flats in a multi-storey building must align across floors to allow smooth flow of water and sewage.

You may have seen engineers referring to drawings at construction sites, tailors sketching patterns for garments or fabricators preparing sketches before making a product – all these are examples of using technical drawings to convey exact information visually.

Technical drawing symbols

Technical drawing is the language of technicians. Like any other language, technical drawing has its rules and grammar. These are represented by 'drawing symbols'. You have already learnt the language of electric circuits – Figure 5.4 shows a few other commonly used symbols. Persons working with machines and materials must learn to use these symbols, and learn the basic concepts of technical drawing to convey ideas.

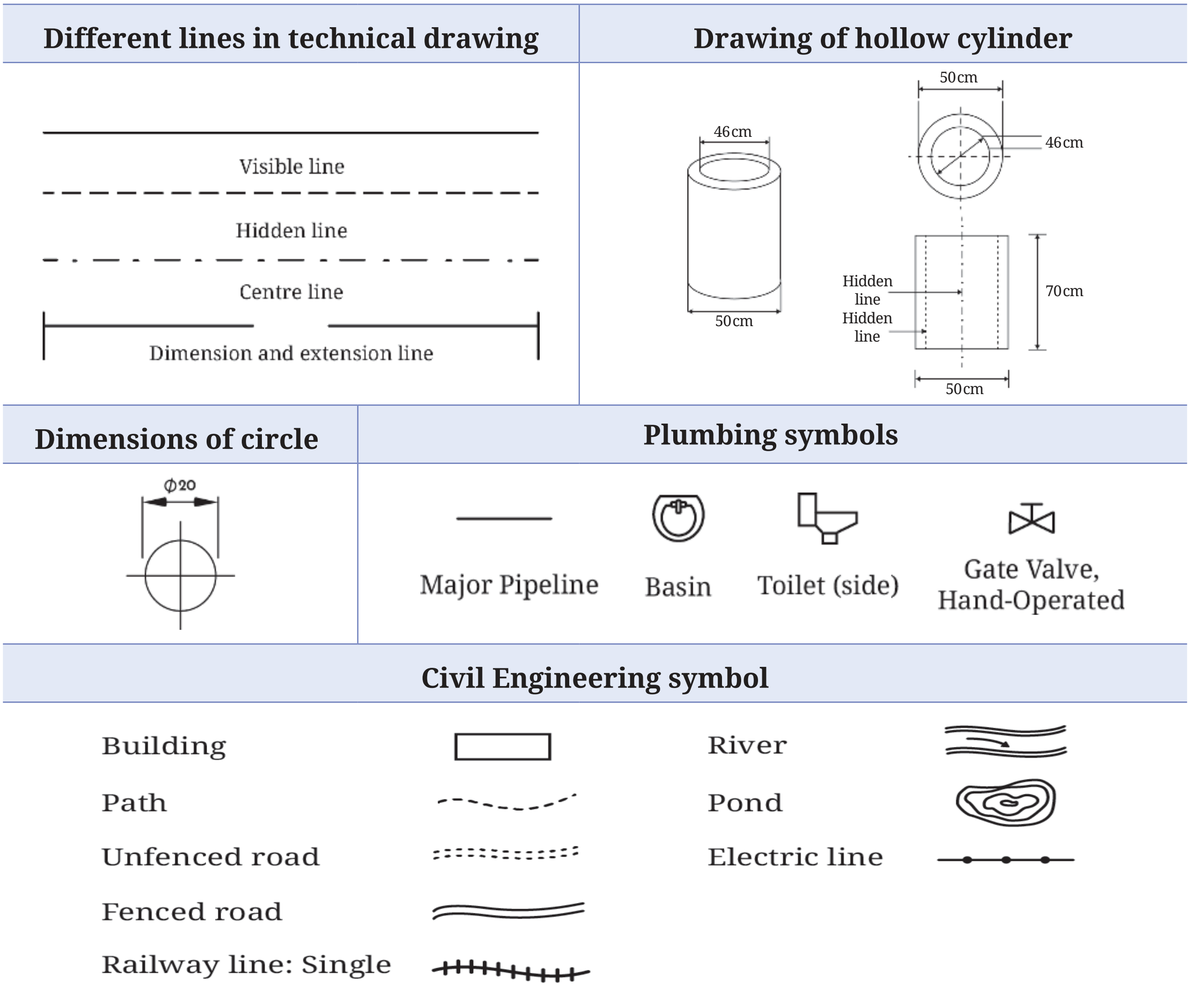


Figure 5.4: *Commonly used lines and symbols in technical drawing*

Drawing an object

Technical drawings must show the exact shape, size and proportion of an object. However, we can see the true dimensions of an object–its actual physical size and shape– only when we look at it from different angles.

For example, look at the table you are working on from the top. Does this give you the complete picture? No, because from this view you cannot see the height of the table. Similarly, from the front view alone, the width may not be visible.

Therefore, to understand the complete shape and size of an object, we need to look at it from different directions–front, top and side. A combination of these views gives a true and complete representation of the object.

Technical drawing

The technique of representing a three-dimensional (3D) object on a two-dimensional (2D) surface using these views is called projection.

Look at the examples of projection of different objects in Figure 5.5 to understand this further.

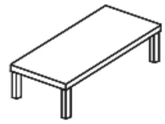
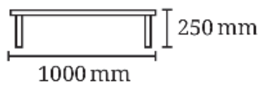
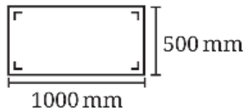
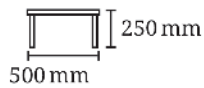
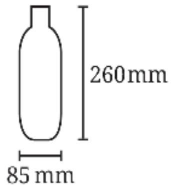
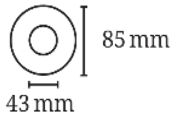
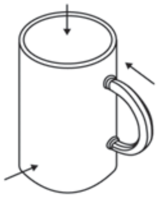
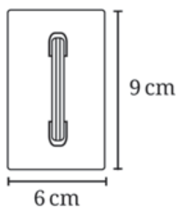
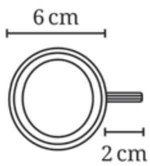
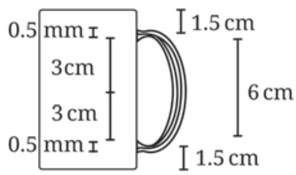
	Front view	Top view	Side view
 Study table			
 Bottle			
 Coffee mug			

Figure 5.5: *Projections of different objects*

In construction, the top view, which shows the details of the 'floor plan' of a building, is called a Plan. The front view is called Elevation. Figure 5.6 shows an example.

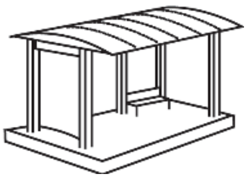
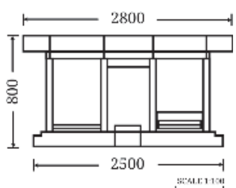
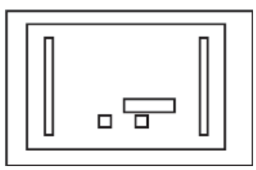
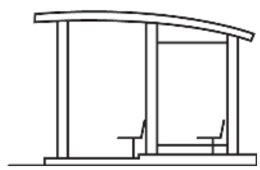
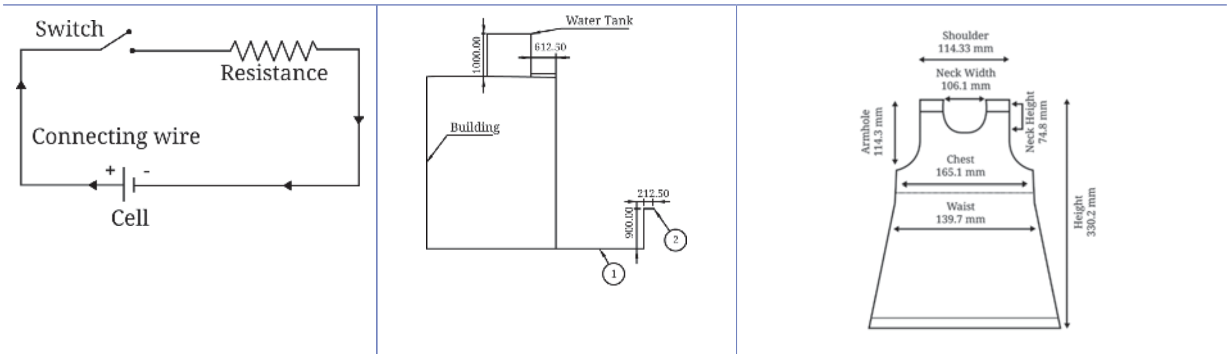
			
3D drawing	Elevation (Front view)	Plan (Top View)	Side view

Figure 5.6: *Technical drawing in construction*

In the case of electric circuits, technical drawings are called circuit diagrams – you are familiar with these from Science. Similarly, a technical drawing showing a plumbing system or the outline of a dress are called a plumbing diagram and pattern, respectively (Figure 5.7).



Typical electric circuit	Plumbing diagram	Technical drawings of a pattern

Figure 5.7: *Technical drawings of electric circuits, plumbing systems and dress pattern*

Scale of drawing

It is not possible to create a technical drawing with the actual dimensions. Try to imagine drawing a full-sized chair on a piece of paper. Therefore, we need to choose a scale to reduce or enlarge the dimensions of an object on paper in proportion to the exact dimensions. The scale is indicated as 1: XX – this means that 1 unit on the drawing corresponds to XX units in real life. Table 5.6 contains a few examples of selecting the scale for different dimensions.

While reading a drawing, you need to refer to the scale mentioned. It is usually written at the bottom of the drawing, in the title box.

Reading and making technical drawings

TASK

Read and draw

Table 5.6 *contains a few examples – look around and add more.*

Table 5.6: Deciding scale of measurement

Actual dimension of object	Scale	Dimensions on the technical drawing		
Height of a shop is 6 m	1:100	1 unit of measurement on drawing is equal to 100 units on ground	100 m = 1 m; 6 m = ? = 1 × 6 / 100 = 0.06 m = 6 cm	6 cm

Length of field on ground is 400 m	1:1000	1 unit of measurement on drawing is equal to 1000 units on ground	1000 m = 1m; 400 m = ? = 1 × 400/1000 = 0.4 m = 40 cm	40 cm
Height of water bottle is 20 cm	1:10	1 unit of measurement on drawing is equal to 10 units in actual	10 cm =1; 20 cm = ? = 1 × 20/10 = 2 cm	2 cm

Now that you have learnt the basics of technical drawing – symbols, dimensions, and scale interpretation – you must apply this knowledge to both read and draw technical drawings.

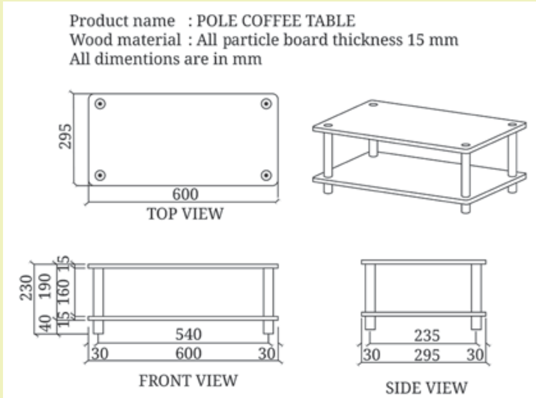


Figure 5.8: *Technical drawing of a table*

Reading technical drawings

Look carefully at Figures 5.8 and 5.9 and answer the questions.

1. What is the total length and breadth of the object?
2. State the dimensions of the top of the object.
3. How many legs does each table have and what is the length of each leg?
4. What is the distance between two legs?

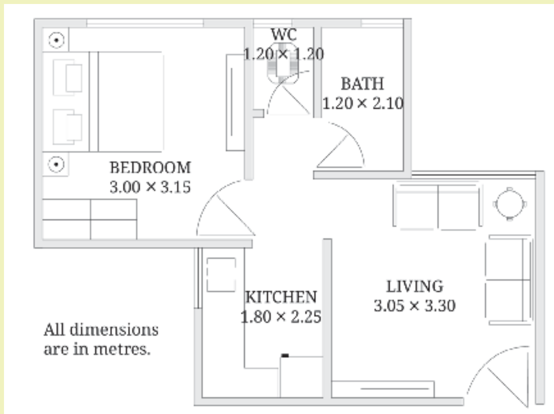


Figure 5.9: *Plan of a house*

Reading the plan of a house

1. What are the dimensions of the kitchen?
2. How many windows are there in the house?
3. How many doors are there in the house?

Creating a technical drawing

On the basis of the instructions and the 3D image in Figure 5.10, draw a stool with accurate measurements.

You need to draw a stool whose actual measurements are:

1. Height – 50 cm, Seat width – 30 cm, Seat depth – 30 cm
2. You can use the scale of 1:10.

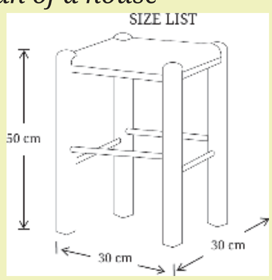


Figure 5.10: *3D drawing of a stool*

5.5 Selection of vocation

This section will help you to decide the vocation you will take up related to work with materials. The textbook provides details of construction and apparel, as well guidelines for sheet metal work, plumbing, food processing, furniture making and pottery. But before that you need to explore vocations around yourself.

TASK

Look around you and answer the following questions:

Explore different sources of information

1. Exploring vocations around us

Look around you and answer the following questions:

a. What kind of work related to shaping materials do you observe around yourself? You can use different sources of information, for example, experts, site visits, libraries, online resources, specific government sites or reports.

b. Briefly describe what the work involves? What are the inputs? What are the key processes and what are the outcomes?

2. Mapping resources

Mapping resources

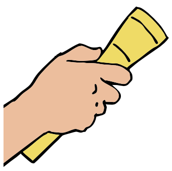
Do you think it is possible for you to do the work in school? Table 5.7 below will help you decide.

Table 5.7: Mapping resource available to do work

Work related to making products from materials	Will you be able to complete the work in the time allocated?	Do you have adequate space to build structures (a wall)?	Have you identified an expert to help you?	Will you be able to collect the tools, materials and machines needed to complete the work?	Will you be able to do 'real' hands-on work?

5.6 When working with materials

1. Selection of material and manufacturing process depends on the characteristics of materials and the desired product.
2. Technical drawings help with the design of the product as well as communication with others who may be involved in developing the product.
3. Tools for measurement should be selected based on what is to be measured and to how much accuracy.
4. Safety is very important – take precautions as per the work you are doing.
5. Use safety signs and symbols wherever needed. There are common symbols and signs for communicating potential risks, and the precautions to be taken.
6. Keep the workplace clean and organised to avoid accidents.



5.7 Assess your learning

1. You are given clay and wood to make a pen stand. Which one of the two (clay and wood) will you choose? Compare the characteristics and explain your decision.
2. Create a safety symbol to caution people about extremely hot surfaces. Think about the colour and image while you design it.
3. Your teacher gives you three objects to measure – a pipe's inner diameter, a cloth length, and the length of the classroom. Which instrument will you use for each and why?
4. Your group made a wooden tray, but it looks uneven and does not stand flat. On the basis of the common steps of developing a product discussed in the chapter, identify what could be the cause of the error.
5. Create a technical drawing (with front, top, and side views) of a simple rack for storing sports items.
6. Of the tasks that you did, which did you enjoy the most? Which did you enjoy the least? Give examples of what went well and what did not go well. What would you do differently next time?
7. Give examples of how you can apply your learnings in a real-life situation.